

Questions — The Innovation Ecosystem

Analytical Questions

1. How does the concept of a hierarchical generative model apply to the structure of an innovation ecosystem? Describe at least three levels of the hierarchy and explain how prediction errors propagate between them.
2. Compare the Markov blanket of a startup entering a new market with the Markov blanket of an established corporation defending its market position. How do the sensory and active states differ in scope, precision, and strategic importance?
3. Explain why platform ecosystems tend toward winner-take-all outcomes using the Active Inference framework. What role does free energy minimization play in the positive feedback loops that drive platform dominance?
4. Analyze how regulatory frameworks function as shared generative models. What happens to an innovation ecosystem when the regulatory model undergoes rapid revision, as occurred during the early regulation of ride-sharing services?
5. The pharmaceutical industry's FDA approval process can be described as a formalized inference procedure. Map each clinical trial phase (I, II, III, IV) to an Active Inference concept (prior, likelihood, posterior, model evidence). What does this mapping reveal about the logic of drug regulation?
6. Distinguish between the free energy landscape of an invention in an emerging ecosystem versus a mature ecosystem. How do the available strategies for free energy minimization differ in each phase?
7. How do standards bodies reduce uncertainty in innovation ecosystems? Discuss the trade-off between the coordination benefits of standards and the constraints they impose on inventive freedom, using a specific example.

8. Analyze the VHS vs. Betamax competition as a multi-agent free energy minimization problem. Why did the technically inferior format prevail, and what does this tell us about the relationship between technical quality and ecosystem fitness?
9. How does Clayton Christensen's disruption theory map onto Active Inference? Explain why incumbent firms' well-optimized generative models make them vulnerable to disruptions that change the structure of the model itself.
10. Describe how an invention's Markov blanket expands as it scales from a prototype to a global product. What new sensory and active states must be developed at each scale, and what are the risks of blanket expansion outpacing organizational capacity?

Applied Questions

1. Choose your invention and identify its complete Markov blanket within its innovation ecosystem. List at least five sensory states, five active states, and five external states. Explain one strategic insight that emerges from this mapping.
2. Your invention operates in an ecosystem that is transitioning from the growth phase to the maturity phase. What specific changes should you expect in the competitive landscape, regulatory environment, and user expectations? How should you adjust your strategy?
3. A dominant platform in your ecosystem announces a new feature that directly competes with your invention. Using Active Inference ecosystem analysis, outline three strategic responses and evaluate the free energy implications of each.
4. You have the opportunity to participate in a standards-setting process relevant to your invention. Describe the potential benefits and risks. How would you determine which standards proposals to support, oppose, or propose yourself?
5. Map the supply chain for your invention as a series of coupled Active Inference systems. Identify at least three points where prediction errors in one link could cascade through the chain. How would you build resilience at these points?

6. Your invention has achieved product-market fit in one geographic market and you are considering expansion to a market with a different regulatory environment. How do you use ecosystem analysis to evaluate the feasibility and plan the expansion?
7. Analyze a real-world ecosystem failure (e.g., Google Glass, Segway, or another example). What ecosystem components were missing, misaligned, or hostile? How could an Active Inference ecosystem analysis have predicted the failure?
8. You are designing a platform strategy for your invention. What affordances must your platform provide to attract complementary innovators? How do you balance openness (attracting developers) with control (maintaining prediction accuracy)?
9. Your invention's ecosystem includes a powerful incumbent that could become either your most important partner or your most dangerous competitor. Using Active Inference, analyze how you would model the incumbent's generative model to predict their response to your invention.
10. Design an "ecosystem health dashboard" for your invention. What five metrics would you track to monitor the health and dynamics of your innovation ecosystem? For each metric, explain what prediction error it would signal and what action you would take in response.